VOICEOPS CAPSTONE PROJECT

ZentroMart Customer Support Voice Agent

Objective	Build a voice-based support prototype that answers customer questions using company policy documents and provides guarded order-support workflows.
Core flow	Customer audio → Gnani STT → VoiceOps agent → retrieval / support tools → Gemini response → Gnani TTS → spoken reply

1. Problem Statement

Customer support requests often require consistent policy answers, order-specific information, and escalation when a request cannot be resolved safely. This project prototypes a voice interface for ZentroMart that combines speech processing, retrieval-augmented generation (RAG), and support tools. Answers should be grounded in the supplied company knowledge base rather than unsupported assumptions.

2. System Architecture

The implementation is organized into five main stages:

- **Speech input:** Gnani Vachana transcribes a customer recording into text using the en-IN locale.
- **Agent orchestration:** the VoiceOps agent routes order-related requests to support logic and other questions to retrieval.
- **Retrieval:** company Markdown files are split into sections and embedded with sentence-transformers/all-MiniLM-L6-v2. Normalized embeddings and cosine similarity rank relevant sections.
- **Answer generation:** Gemini receives the question and retrieved excerpts, with instructions to stay grounded and cite policy section IDs.
- **Speech output:** Gnani Vachana synthesizes the agent reply into audio using the configured voice and language.

3. Technologies

Component	Technology / role
Application	Python; local virtual environment
LLM	Google Gemini API for response generation
Retrieval	Sentence Transformers, all-MiniLM-L6-v2, NumPy cosine similarity
Voice	Gnani Vachana STT and TTS
Support data	Provided CSV datasets and support tools

4. Retrieval and Policy Grounding

Retrieved passages retain metadata such as source file, section ID, status, effective date, and similarity score. For example, a test query about returning an electronic product retrieved RET-3 (electronics return-window change) and RET-4 (return condition requirements). The generated answer distinguished the 10-day window for electronics delivered before 1 August 2026 from the 7-day window for electronics and accessories delivered from that date onward, and cited the relevant sections.

5. Conversation Handling and Security

For order-related requests, the agent asks the customer to verify using either the last four digits of the registered phone number or the six-digit delivery pincode before revealing order details. The support logic limits the verification retry. Ticket creation is reported as successful only when the underlying tool returns a successful creation result. Synthetic support data is used for development and testing.

6. Voice Pipeline Test

An end-to-end test processed a customer recording with Gnani STT, obtained the transcript “hello i want to return an electronic product can i get refund”, generated a policy-grounded response with RET-3 and RET-4 citations, and synthesized reply audio with Gnani TTS. The test log reported successful transcription and synthesis. This confirms a working prototype path for the tested recording, not exhaustive voice quality or reliability.

7. Evaluation and Limitations

The initial evaluation script compares LLM-only answers with RAG answers against required and prohibited phrase checks in practice/questions.csv. The first run encountered frequent API/rate-limit or processing errors: LLM-only had 17 completed cases out of 52, with 6 passing both phrase checks; RAG had 15 completed cases out of 52, with 7 passing both checks.

These are preliminary automated phrase-check results only. They are not a verified accuracy score, and the different numbers of completed cases make a direct performance comparison unreliable. The run does not establish that RAG is better. A reliable evaluation requires rerunning failed cases when API capacity allows, manual review against reference evidence, and systematic multi-turn testing using practice/conversations.csv.

Other limitations include the current retrieval approach relying on embedding similarity rather than a hybrid ranker, limited automated assertions in the existing manual test scripts, and the need for broader checks of tool failures, policy edge cases, multilingual behavior, and audio conditions.

8. Future Work

- Add robust retries and backoff for transient API failures, without repeatedly spending quota during testing.
- Evaluate single-turn questions and multi-turn conversation scenarios separately, including expected tool calls and evidence.
- Review retrieval results for difficult queries and assess whether lexical or hybrid retrieval improves coverage.
- Add automated assertions for verification, escalation, citations, and policy boundaries.

9. Conclusion

The prototype demonstrates an integrated customer-support flow using Gnani speech services, a Gemini-powered response generator, document retrieval, and guarded support tools. Basic retrieval, agent, support-tool, and end-to-end voice checks succeeded. The evaluation remains preliminary and should be strengthened before making quantitative claims about answer quality.

Prepared from the implementation and test outputs described during project development. Results are prototype-level and should be interpreted with the limitations stated above.